

1. Questions

Study the following information carefully and answer the questions given below:

The given table shows the total quantity of vegetables available, the percentage of the quantity of potatoes available out of the total quantity of vegetables available, the ratio of the quantity of onions available to that of carrots, and the percentage of the quantity of vegetables sold out of the total quantity of vegetables available in five different shops i.e. A, B, C, D and E.

Notes:

I). The quantity of vegetables in (Kg).

II). The difference between the quantity of vegetables unsold in shops D and E is 36.

Shops	The total quantity of vegetables available	The percentage of the quantity of potatoes available	The ratio of the quantity of onions available to that of carrots	The percentage of the quantity of vegetables sold
A	180	44.44%	11: 9	(30% of N)%
B	$N + 50$	28%	-	80%
C	150	32%	9: 8	66.66%
D	$2N$	27.5%	7: 3	72.5%
E	N	-	4: 3	63%

If the purchasing price per kg of onion, potato, and carrot in shop A is Rs. $2x$, Rs. $(2x + 10)$, and Rs. x , respectively, and the total purchasing cost in shop A is Rs. 7100, find the value of x .

- 25
- 10
- 20
- 15
- 30

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2. Questions

If the ratio of the quantity of potatoes sold, onions sold, and carrots sold in shop E is 8:3:7, find the sum of the difference between the quantity of onions sold and the quantity of carrots sold in shop E and the quantity of unsold vegetables in shop C

- 78 kg
- 72 kg
- 76 kg
- 68 kg

e. 82 kg

3. Questions

If the quantity of potatoes available in shop E is 92.5% of that in shop A, then find the ratio of the average quantity of carrots available in shops C, D, and E together to the quantity of vegetables unsold in shop A?

- a. 3: 4
- b. 2: 3
- c. 8: 9
- d. 7: 9
- e. 7: 8

4. Questions

If the quantity of carrots available in shops B and E is $(N - 120)$ kg, and $(N - 146)$ kg, respectively, then the difference between the quantity of onions available in shops A and B is how much percentage of the sum of the quantity of carrots available in shop C and the quantity of onions available in shop E?

- a. 32.5%
- b. 27.5%
- c. 42.5%
- d. 37.5%
- e. 22.5%

5. Questions

If the ratio of the quantity of potatoes to that of onions in shop B is 7: 10 respectively. The quantity of potatoes and carrots available in shop B is how much less than the quantity of potatoes and onions available in shop D?

- a. 155 kg
- b. 163 kg
- c. 133 kg
- d. 161 kg
- e. 143 kg

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6. Questions

Study the following information carefully and answer the questions given below:

There are three products available in a shop. The cost price of product B is 6 times that of product C. The

selling price of product A is Rs. 1604 less than the cost price of product B, and the marked-up percentages of products B and C is $2x\%$ and $5y\%$, respectively. The profit percentage of product B is $(x/5)\%$, and the ratio of the selling prices of the products A, B, and C is 1598: 2520: 495. The discount percentage on product A is $x - 10\%$.

Notes:

i). $x + y = 38$

ii). $x^2 - y^2 = 456$

The difference between the discount price on products A and C is how much percentage of the profit price on product B?

- a. 92.5%
- b. 87.5%
- c. 97.5%
- d. 82.5%
- e. 85.5%

7. Questions

If the product B is sold at two successive discounts of $y + 7\%$ and $z\%$ respectively, which of the following statement(s) is/are true?

I). $x = 2z$

II). $y = z$

III). $y = 2z + 0.5$

- a. Only I
- b. Only II
- c. Only III
- d. None of them
- e. All I, II, and III

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8. Questions

If the profit percentage on product A is 59.8%, then find the ratio of marked-up percentage on product A to that on product B.

- a. 8: 5
- b. 44: 25
- c. 6: 5
- d. 7: 10

e. 5: 1

9. Questions

If the cost price of product D is Rs. $(xy - 125)$ more than that of product B, and the ratio of the marked-up percentage on products C and D is 5:6 respectively, while the ratio of the discount percentage on products C and D is 5:8 respectively, find the average selling price of products B, C, and D.

- a. Rs. 3360
- b. Rs. 3630
- c. Rs. 3330
- d. Rs. 3790
- e. Rs. 3390

10. Questions

If the cost price of product A is three times that of product C, find the sum of the profit prices on products A, B, and C together.

- a. Rs. 1276
- b. Rs. 1146
- c. Rs. 1056
- d. Rs. 1796
- e. Rs. 1226

11. Questions

The following questions have two quantities as Quantity I and Quantity II. You have to determine the relationship between them and give answer as,

Both 'x' and 'y' are integers.

Quantity I: Find the value of $(343x^3)^{(1/3)} - 3x^4 + 67x$, such that $3 \geq x \geq 2$.

Quantity II: Find the value of $24 * (729)^{(1/y)} - 31y$, such that $4 > y \geq 2$

- a. Quantity I > Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity I < Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II or Relation cannot be established

12. Questions

The area of the equilateral triangle is $(\sqrt{3}d^2/4) \text{ m}^2$. The triangle is drawn on a rectangle with an area of 234 m^2 , where the difference between the length and breadth of the rectangle is 5 m. The breadth of the rectangle is equal to the side of the equilateral triangle.

Quantity I: Find the value of ' d^2 '

Quantity II: If the side of the square is 200% more than the value of ' d '. Find the area of square.

- a. Quantity I < Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity I > Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II or Relation cannot be established

13. Questions

A thief escaped from jail and ran at a speed of 43.2 km/h. The police noticed the thief after $66\frac{2}{3}$ seconds and caught him after 1 minute and 40 seconds. The speed of the police is ' s ' m/sec.

Quantity I: If the speed of the stream is 4 km/h and the speed of the boat is ' s ' m/sec, find the time taken by the boat to travel 380 km downstream.

Quantity II: Find the time taken by the police to cover a distance of 360 km.

- a. Quantity I > Quantity II
- b. Quantity I = Quantity II or Relation cannot be established
- c. Quantity I < Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I $\geq$ Quantity II

14. Questions

The ratio of A's age, 8 years ago to C's age, 5 years ago is 2:5. The ratio of the present ages of A to B is 1:3 and B's age 6 years from now will be twice C's age 10 years ago. The present ages of A, B, and C are ' x ', ' y ', and ' z ', respectively.

Quantity I: If the average age of B, C, and D is ' $z + 2$ ' years, find the difference between the present age of D and A's age 6 years ago.

Quantity II: Find the value of ' $(x + z) * 10 \div y$ '.

- a. Quantity I < Quantity II
- b. Quantity I = Quantity II or Relation cannot be established
- c. Quantity I > Quantity II

d. Quantity I $\leq$ Quantity II

e. Quantity I $\geq$ Quantity II

15. Questions

The ratio of the difference between the simple interest (SI) and compound interest (CI) for 2 years to the difference between the simple interest and compound interest for 3 years on the same principal is 10:31, respectively. If the simple interest earned on the principal (Rs. P) for a year is Rs. 2500.

Quantity I: If Rs. (P + 2250) is invested in Scheme A at 9% simple interest per annum, find the interest earned after 4 years.

Quantity II: If Rs. (P – 5000) is invested in Scheme B at 20% per annum, compounded annually, find the interest earned after 3 years.

a. Quantity I < Quantity II

b. Quantity I $\geq$ Quantity II

c. Quantity I > Quantity II

d. Quantity I $\leq$ Quantity II

e. Quantity I = Quantity II or Relation cannot be established

16. Questions

There is a wrong term and the correct term of the series is p and q, respectively. Find the nearest possible value of 'p + q'?

Series: 243, 486, 162, 327, 108, 216, 72

a. 657

b. 427

c. 648

d. 148

e. 655

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17. Questions

Study the following information carefully and answer the following question.

Series: (X – 7), X, 76, 95, 123, 162

If the another series is started with $\sqrt{X}$ and it follows the same pattern as the given series, then find the $(\sqrt[3]{X})^{\text{th}}$ term of the new series.

a. 46

b. 27

- c. 74
- d. 15
- e. None of the above

18. Questions

The value of 'y' is $(X - 14)\%$ more than the value of X. If the first term of the series is decreased by 3, what percentage does it become of the value of 'y'?

- a. 56.75%
- b. 56.5%
- c. 54.5%
- d. 57.25%
- e. 56.25%

19. Questions

Study the following information carefully and answer the following question.

The following two series contain the variables 'p' and 'q,' and each series follows a different pattern.

Series I: $500q$, 2500, $(q - 25)$, p, 0.16

Series II: $500q$, 625, 12.5, 0.5, 0.04

Find the value of ' $q/(p + 1)$ '?

- a. 62.5
- b. 12.5
- c. 31.25
- d. 25
- e. 50

20. Questions

Find the missing term of the following series.

Series: p, 2p, 6p, ?, $(q + 355)$, 720p

- a. 80
- b. 88
- c. 96
- d. 100

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e. 112

Explanations:

1. Questions

The quantity of vegetables unsold in shop D = $(100 - 72.5)\%$ of $2N = (27.5/100) * 2N = 0.55N$

The quantity of vegetables unsold in shop E = $(100 - 63)\%$ of $N = 0.37N$

According to the Note II,

$$0.55N - 0.37N = 36$$

$$0.18N = 36$$

$$N = 200$$

In shop D,

The total quantity of vegetables available = $2N = 400$ kg

The quantity of potatoes available = $(27.5/100) * 400 = 110$ kg

The sum of the quantity of onions and carrots available = $400 - 110 = 290$ kg

The quantity of onions available = $290 * (7/10) = 203$ kg

The quantity of carrots available = $290 - 203 = 87$ kg

The quantity of vegetables sold = $(72.5/100) * 400 = 290$ kg

The quantity of vegetables unsold = $400 - 290 = 110$ kg

In shop E,

The total quantity of vegetables available = $N = 200$ kg

The quantity of vegetables sold = $(63/100) * 200 = 126$ kg

The quantity of vegetables unsold = $200 - 126 = 74$ kg

In shop A,

The total quantity of vegetables available = 180 kg

The quantity of potatoes available = 44.44% of 180 = $(4/9) * 180 = 80$ kg

The sum of the quantity of onions and carrots available = $180 - 80 = 100$ kg

The quantity of onions available = $100 * (11/20) = 55$ kg

The quantity of carrots available = $100 - 55 = 45$ kg

The quantity of vegetables sold = $(30\% \text{ of } 200)\%$ of 180 = $(60/100) * 180 = 108$ kg

The quantity of vegetables unsold = $180 - 108 = 72$ kg

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In shop B,

The total quantity of vegetables available = $N + 50 = 200 + 50 = 250$ kg

The quantity of potatoes available = $(28/100)$ of $250 = 70$ kg

The sum of the quantity of onions and carrots available = $250 - 70 = 180$ kg

The quantity of vegetables sold = $(80/100) * 250 = 200$ kg

The quantity of vegetables unsold = $250 - 200 = 50$ kg

In shop C,

The total quantity of vegetables available = 150 kg

The quantity of potatoes available = $(32/100) * 150 = 48$ kg

The sum of the quantity of onions and carrots available = $150 - 48 = 102$ kg

The quantity of onions available = $102 * (9/17) = 54$ kg

The quantity of carrots available = $102 - 54 = 48$ kg

The quantity of vegetables sold = 66.66% of $150 = (2/3) * 150 = 100$ kg

The quantity of vegetables unsold = $150 - 100 = 50$ kg

Let table,

Shops	The total quantity of vegetables available	The quantity of potatoes available	The quantity of onions available	The quantity of carrots available	The quantity of vegetables sold	The quantity of vegetables unsold
A	180	80	55	45	108	72
B	250	70	-	-	200	50
C	150	48	54	48	100	50
D	400	110	203	87	290	110
E	200	-	-	-	126	74

Answer: C

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According to the question,

$$(55 * 2x) + [80 * (2x + 10)] + (45 * x) = 7100$$

$$110x + 160x + 800 + 45x = 7100$$

$$315x = 6300$$

$$x = 20$$

2. Questions

The quantity of vegetables unsold in shop D = $(100 - 72.5)\%$ of $2N = (27.5/100) * 2N = 0.55N$

The quantity of vegetables unsold in shop E = $(100 - 63)\%$ of $N = 0.37N$

According to the Note II,

$$0.55N - 0.37N = 36$$

$$0.18N = 36$$

$$N = 200$$

In shop D,

The total quantity of vegetables available = $2N = 400$ kg

The quantity of potatoes available = $(27.5/100) * 400 = 110$ kg

The sum of the quantity of onions and carrots available = $400 - 110 = 290$ kg

The quantity of onions available = $290 * (7/10) = 203$ kg

The quantity of carrots available = $290 - 203 = 87$ kg

The quantity of vegetables sold = $(72.5/100) * 400 = 290$ kg

The quantity of vegetables unsold = $400 - 290 = 110$ kg

In shop E,

The total quantity of vegetables available = $N = 200$ kg

The quantity of vegetables sold = $(63/100) * 200 = 126$ kg

The quantity of vegetables unsold = $200 - 126 = 74$ kg

In shop A,

The total quantity of vegetables available = 180 kg

The quantity of potatoes available = 44.44% of $180 = (4/9) * 180 = 80$ kg

The sum of the quantity of onions and carrots available = $180 - 80 = 100$ kg

The quantity of onions available = $100 * (11/20) = 55$ kg

The quantity of carrots available = $100 - 55 = 45$ kg

The quantity of vegetables sold = $(30\% \text{ of } 200)\% \text{ of } 180 = (60/100) * 180 = 108$ kg

The quantity of vegetables unsold = $180 - 108 = 72$ kg

In shop B,

The total quantity of vegetables available = $N + 50 = 200 + 50 = 250$ kg

The quantity of potatoes available = $(28/100)$ of $250 = 70$ kg

The sum of the quantity of onions and carrots available = $250 - 70 = 180$ kg

The quantity of vegetables sold = $(80/100) * 250 = 200$ kg

The quantity of vegetables unsold = $250 - 200 = 50$ kg

In shop C,

The total quantity of vegetables available = 150 kg

The quantity of potatoes available = $(32/100) * 150 = 48$ kg

The sum of the quantity of onions and carrots available = $150 - 48 = 102$ kg

The quantity of onions available = $102 * (9/17) = 54$ kg

The quantity of carrots available = $102 - 54 = 48$ kg

The quantity of vegetables sold = 66.66% of 150 = $(2/3) * 150 = 100$ kg

The quantity of vegetables unsold = $150 - 100 = 50$ kg

Let table,

Shops	The total quantity of vegetables available	The quantity of potatoes available	The quantity of onions available	The quantity of carrots available	The quantity of vegetables sold	The quantity of vegetables unsold
A	180	80	55	45	108	72
B	250	70	-	-	200	50
C	150	48	54	48	100	50
D	400	110	203	87	290	110
E	200	-	-	-	126	74

Answer: A

The difference between the quantity of onions sold and the quantity of carrots sold in shop E = $126 * \{(7 - 3)/18\} = 7 * 4 = 28$ kg

Thus, Sum = $28 + 50 = 78$ kg

3. Questions

The quantity of vegetables unsold in shop D = $(100 - 72.5)\%$ of $2N = (27.5/100) * 2N = 0.55N$

The quantity of vegetables unsold in shop E = $(100 - 63)\%$ of $N = 0.37N$

According to the Note II,

$$0.55N - 0.37N = 36$$

$$0.18N = 36$$

$$N = 200$$

In shop D,

The total quantity of vegetables available = $2N = 400$ kg

The quantity of potatoes available = $(27.5/100) * 400 = 110$ kg

The sum of the quantity of onions and carrots available = $400 - 110 = 290$ kg

The quantity of onions available = $290 * (7/10) = 203$ kg

The quantity of carrots available = $290 - 203 = 87$ kg

The quantity of vegetables sold = $(72.5/100) * 400 = 290$ kg

The quantity of vegetables unsold = $400 - 290 = 110$ kg

In shop E,

The total quantity of vegetables available = $N = 200$ kg

The quantity of vegetables sold = $(63/100) * 200 = 126$ kg

The quantity of vegetables unsold = $200 - 126 = 74$ kg

In shop A,

The total quantity of vegetables available = 180 kg

The quantity of potatoes available = 44.44% of $180 = (4/9) * 180 = 80$ kg

The sum of the quantity of onions and carrots available = $180 - 80 = 100$ kg

The quantity of onions available = $100 * (11/20) = 55$ kg

The quantity of carrots available = $100 - 55 = 45$ kg

The quantity of vegetables sold = $(30\% \text{ of } 200)\% \text{ of } 180 = (60/100) * 180 = 108$ kg

The quantity of vegetables unsold = $180 - 108 = 72$ kg

In shop B,

The total quantity of vegetables available = $N + 50 = 200 + 50 = 250$ kg

The quantity of potatoes available = $(28/100)$ of $250 = 70$ kg

The sum of the quantity of onions and carrots available = $250 - 70 = 180$ kg

The quantity of vegetables sold = $(80/100) * 250 = 200$ kg

The quantity of vegetables unsold = $250 - 200 = 50$ kg

In shop C,

The total quantity of vegetables available = 150 kg

The quantity of potatoes available = $(32/100) * 150 = 48$ kg

The sum of the quantity of onions and carrots available = $150 - 48 = 102$ kg

The quantity of onions available = $102 * (9/17) = 54$ kg

The quantity of carrots available = $102 - 54 = 48$ kg

The quantity of vegetables sold = $66.66\% \text{ of } 150 = (2/3) * 150 = 100$ kg

The quantity of vegetables unsold = $150 - 100 = 50$ kg

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Let table,

Shops	The total quantity of vegetables available	The quantity of potatoes available	The quantity of onions available	The quantity of carrots available	The quantity of vegetables sold	The quantity of vegetables unsold
A	180	80	55	45	108	72
B	250	70	-	-	200	50
C	150	48	54	48	100	50
D	400	110	203	87	290	110
E	200	-	-	-	126	74

Answer: E

The quantity of potatoes available in shop E = $(92.5/100) * 80 = 74$ kg

The sum of the quantity of onions and carrots available in shop E = $200 - 74 = 126$ kg

The quantity of onions available in shop E = $126 * (3/7) = 54$ kg

The average quantity of carrots available in shops C, D, and E together = $(48 + 87 + 54)/3 = 189/3 = 63$

Thus, ratio = $63: 72 = 7: 8$

4. Questions

The quantity of vegetables unsold in shop D = $(100 - 72.5)\%$ of $2N = (27.5/100) * 2N = 0.55N$

The quantity of vegetables unsold in shop E = $(100 - 63)\%$ of $N = 0.37N$

According to the Note II,

$$0.55N - 0.37N = 36$$

$$0.18N = 36$$

$$N = 200$$

In shop D,

The total quantity of vegetables available = $2N = 400$ kg

The quantity of potatoes available = $(27.5/100) * 400 = 110$ kg

The sum of the quantity of onions and carrots available = $400 - 110 = 290$ kg

The quantity of onions available = $290 * (7/10) = 203$ kg

The quantity of carrots available = $290 - 203 = 87$ kg

The quantity of vegetables sold = $(72.5/100) * 400 = 290$ kg

The quantity of vegetables unsold = $400 - 290 = 110$ kg

In shop E,

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The total quantity of vegetables available = $N = 200$ kg

The quantity of vegetables sold = $(63/100) * 200 = 126$ kg

The quantity of vegetables unsold = $200 - 126 = 74$ kg

In shop A,

The total quantity of vegetables available = 180 kg

The quantity of potatoes available = 44.44% of 180 = $(4/9) * 180 = 80$ kg

The sum of the quantity of onions and carrots available = $180 - 80 = 100$ kg

The quantity of onions available = $100 * (11/20) = 55$ kg

The quantity of carrots available = $100 - 55 = 45$ kg

The quantity of vegetables sold = (30% of 200)% of 180 = $(60/100) * 180 = 108$ kg

The quantity of vegetables unsold = $180 - 108 = 72$ kg

In shop B,

The total quantity of vegetables available = $N + 50 = 200 + 50 = 250$ kg

The quantity of potatoes available = $(28/100)$ of 250 = 70 kg

The sum of the quantity of onions and carrots available = $250 - 70 = 180$ kg

The quantity of vegetables sold = $(80/100) * 250 = 200$ kg

The quantity of vegetables unsold = $250 - 200 = 50$ kg

In shop C,

The total quantity of vegetables available = 150 kg

The quantity of potatoes available = $(32/100) * 150 = 48$ kg

The sum of the quantity of onions and carrots available = $150 - 48 = 102$ kg

The quantity of onions available = $102 * (9/17) = 54$ kg

The quantity of carrots available = $102 - 54 = 48$ kg

The quantity of vegetables sold = 66.66% of 150 = $(2/3) * 150 = 100$ kg

The quantity of vegetables unsold = $150 - 100 = 50$ kg

Let table,

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Shops	The total quantity of vegetables available	The quantity of potatoes available	The quantity of onions available	The quantity of carrots available	The quantity of vegetables sold	The quantity of vegetables unsold
A	180	80	55	45	108	72
B	250	70	-	-	200	50
C	150	48	54	48	100	50
D	400	110	203	87	290	110
E	200	-	-	-	126	74

Answer: D

The quantity of carrots available in shop B = $N - 120 = 200 - 120 = 80$ kg

The quantity of onions available in shop B = $180 - 80 = 100$ kg

The difference between the quantity of onions available in shops A and B = $100 - 55 = 45$ kg

The quantity of carrots available in shop E = $N - 146 = 200 - 146 = 54$ kg

The quantity of onions available in shop E = $54 * (4/3) = 72$ kg

The sum of the quantity of carrots available in shop C and the quantity of onions available in shop E = $48 + 72 = 120$ kg

Thus, percentage = $(45/120) * 100 = 37.5\%$

5. Questions

The quantity of vegetables unsold in shop D = $(100 - 72.5)\%$ of $2N = (27.5/100) * 2N = 0.55N$

The quantity of vegetables unsold in shop E = $(100 - 63)\%$ of $N = 0.37N$

According to the Note II,

$$0.55N - 0.37N = 36$$

$$0.18N = 36$$

$$N = 200$$

In shop D,

The total quantity of vegetables available = $2N = 400$ kg

The quantity of potatoes available = $(27.5/100) * 400 = 110$ kg

The sum of the quantity of onions and carrots available = $400 - 110 = 290$ kg

The quantity of onions available = $290 * (7/10) = 203$ kg

The quantity of carrots available = $290 - 203 = 87$ kg

The quantity of vegetables sold = $(72.5/100) * 400 = 290$ kg

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The quantity of vegetables unsold = $400 - 290 = 110$ kg

In shop E,

The total quantity of vegetables available = $N = 200$ kg

The quantity of vegetables sold = $(63/100) * 200 = 126$ kg

The quantity of vegetables unsold = $200 - 126 = 74$ kg

In shop A,

The total quantity of vegetables available = 180 kg

The quantity of potatoes available = 44.44% of $180 = (4/9) * 180 = 80$ kg

The sum of the quantity of onions and carrots available = $180 - 80 = 100$ kg

The quantity of onions available = $100 * (11/20) = 55$ kg

The quantity of carrots available = $100 - 55 = 45$ kg

The quantity of vegetables sold = $(30\% \text{ of } 200)\% \text{ of } 180 = (60/100) * 180 = 108$ kg

The quantity of vegetables unsold = $180 - 108 = 72$ kg

In shop B,

The total quantity of vegetables available = $N + 50 = 200 + 50 = 250$ kg

The quantity of potatoes available = $(28/100)$ of $250 = 70$ kg

The sum of the quantity of onions and carrots available = $250 - 70 = 180$ kg

The quantity of vegetables sold = $(80/100) * 250 = 200$ kg

The quantity of vegetables unsold = $250 - 200 = 50$ kg

In shop C,

The total quantity of vegetables available = 150 kg

The quantity of potatoes available = $(32/100) * 150 = 48$ kg

The sum of the quantity of onions and carrots available = $150 - 48 = 102$ kg

The quantity of onions available = $102 * (9/17) = 54$ kg

The quantity of carrots available = $102 - 54 = 48$ kg

The quantity of vegetables sold = $66.66\% \text{ of } 150 = (2/3) * 150 = 100$ kg

The quantity of vegetables unsold = $150 - 100 = 50$ kg

Let table,

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Shops	The total quantity of vegetables available	The quantity of potatoes available	The quantity of onions available	The quantity of carrots available	The quantity of vegetables sold	The quantity of vegetables unsold
A	180	80	55	45	108	72
B	250	70	-	-	200	50
C	150	48	54	48	100	50
D	400	110	203	87	290	110
E	200	-	-	-	126	74

Answer: B

The quantity of onions available in shop B = $70 * (10/7) = 100$ kg

The quantity of carrots available in shop B = $180 - 100 = 80$ kg

The quantity of potatoes and carrots available in shop B = $70 + 80 = 150$ kg

The quantity of potatoes and onions available in shop D = $110 + 203 = 313$ kg

Thus, difference = $313 - 150 = 163$ kg

6. Questions

According to the Note,

$$x + y = 38 \rightarrow (1)$$

$$x^2 - y^2 = 456$$

$$(x + y)(x - y) = 456$$

$$38 * (x - y) = 456$$

$$x - y = 12 \rightarrow (2)$$

Solve (1) and (2)

$$2x = 50$$

$$x = 25$$

and

$$y = 38 - 25$$

$$y = 13$$

Let the cost price of product C be Rs. c

The cost price of product B = $6 * c = \text{Rs. } 6c$

The selling price of product A = Rs. $(6c - 1604)$

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The profit percentage on product B = $(x/5)\% = (25/5)\% = 5\%$

The selling price of product B = $(105/100) * 6c = 6.3c$

The ratio of the selling prices of products A, B, and C is 1598: 2520: 495

According to the data,

$$(6c - 1604)/6.3c = 1598/2520$$

$$(6c - 1604)/c = 799/200$$

$$1200c - 320800 = 799c$$

$$401c = 320800$$

$$c = 800$$

Then,

The cost price of product C = $c = \text{Rs. } 800$

The cost price of product B = $6 * 800 = \text{Rs. } 4800$

The selling price of product A = $4800 - 1604 = \text{Rs. } 3196$

The selling price of product B = $6.3 * 800 = \text{Rs. } 5040$

The selling price of product C = $5040 * (495/2520) = \text{Rs. } 990$

The discount percentage on product A = $(x - 10)\% = 15\%$

The marked price on product A = $3196 * (100/85) = \text{Rs. } 3760$

The marked-up percentage of products B and C is '2x'% and '5y'% , respectively

So, the marked-up percentage of the products B and C is 50% and 65%, respectively.

The marked price on product B = $(150/100)* 4800 = \text{Rs. } 7200$

The marked price on product C = 165% of 800 = Rs. 1320

The profit price on product B = $5040 - 4800 = \text{Rs. } 240$

The profit price on product C = $990 - 800 = \text{Rs. } 190$

The discount price on product A = $3760 - 3196 = \text{Rs. } 564$

The discount price on product B = $7200 - 5040 = \text{Rs. } 2160$

The discount price on product C = $1320 - 990 = \text{Rs. } 330$

Let table,

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Products	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	-	3760	3196	-	564
B	4800	7200	5040	240	2160
C	800	1320	990	190	330

Answer: C

The difference between the discount price on products A and C = $564 - 330 = \text{Rs. } 234$

Thus, percentage = $(234/240) * 100 = 97.5\%$

7. Questions

According to the Note,

$$x + y = 38 \rightarrow (1)$$

$$x^2 - y^2 = 456$$

$$(x + y)(x - y) = 456$$

$$38 * (x - y) = 456$$

$$x - y = 12 \rightarrow (2)$$

Solve (1) and (2)

$$2x = 50$$

$$x = 25$$

and

$$y = 38 - 25$$

$$y = 13$$

Let the cost price of product C be Rs. c

The cost price of product B = $6 * c = \text{Rs. } 6c$

The selling price of product A = Rs. $(6c - 1604)$

The profit percentage on product B = $(x/5)\% = (25/5)\% = 5\%$

The selling price of product B = $(105/100) * 6c = 6.3c$

The ratio of the selling prices of products A, B, and C is 1598: 2520: 495

According to the data,

$$(6c - 1604)/6.3c = 1598/2520$$

$$(6c - 1604)/c = 799/200$$

$$1200c - 320800 = 799c$$

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$$401c = 320800$$

$$c = 800$$

Then,

The cost price of product C = $c = \text{Rs. } 800$

The cost price of product B = $6 * 800 = \text{Rs. } 4800$

The selling price of product A = $4800 - 1604 = \text{Rs. } 3196$

The selling price of product B = $6.3 * 800 = \text{Rs. } 5040$

The selling price of product C = $5040 * (495/2520) = \text{Rs. } 990$

The discount percentage on product A = $(x - 10)\% = 15\%$

The marked price on product A = $3196 * (100/85) = \text{Rs. } 3760$

The marked-up percentage of products B and C is '2x%' and '5y%', respectively

So, the marked-up percentage of the products B and C is 50% and 65%, respectively.

The marked price on product B = $(150/100) * 4800 = \text{Rs. } 7200$

The marked price on product C = 165% of 800 = Rs. 1320

The profit price on product B = $5040 - 4800 = \text{Rs. } 240$

The profit price on product C = $990 - 800 = \text{Rs. } 190$

The discount price on product A = $3760 - 3196 = \text{Rs. } 564$

The discount price on product B = $7200 - 5040 = \text{Rs. } 2160$

The discount price on product C = $1320 - 990 = \text{Rs. } 330$

Let table,

Products	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	-	3760	3196	-	564
B	4800	7200	5040	240	2160
C	800	1320	990	190	330

Answer: A

The selling price of product B = Rs. 5040

The marked price on product B = Rs. 7200

The two successive discounts 20% and z% (Here y = 13)

Let,

$$7200 * (80/100) * (100 - z)/100 = 5040$$

$$100 - z = 5040/57.6$$

$$z = 100 - 87.5$$

$$z = 12.5$$

$$\text{WKT, } x = 25$$

$$\text{I). } x = 2z$$

Let,

$$25 = 2 * 12.5$$

$$25 = 25$$

Statement I is satisfied.

$$\text{II). } y = z$$

Let,

$$13 \neq 12.5$$

Statement II is not satisfied

$$\text{III). } y = 2z + 0.5$$

Let,

$$13 = 25 + 0.5$$

$$13 \neq 25.5$$

Statement III is not satisfied

So, Statement I is true.

Hence, option A

8. Questions

According to the Note,

$$x + y = 38 \rightarrow (1)$$

$$x^2 - y^2 = 456$$

$$(x + y)(x - y) = 456$$

$$38 * (x - y) = 456$$

$$x - y = 12 \rightarrow (2)$$

Solve (1) and (2)

$$2x = 50$$

$$x = 25$$

and

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$$y = 38 - 25$$

$$y = 13$$

Let the cost price of product C be Rs. c

The cost price of product B = $6 * c = \text{Rs. } 6c$

The selling price of product A = Rs. $(6c - 1604)$

The profit percentage on product B = $(x/5)\% = (25/5)\% = 5\%$

The selling price of product B = $(105/100) * 6c = 6.3c$

The ratio of the selling prices of products A, B, and C is 1598: 2520: 495

According to the data,

$$(6c - 1604)/6.3c = 1598/2520$$

$$(6c - 1604)/c = 799/200$$

$$1200c - 320800 = 799c$$

$$401c = 320800$$

$$c = 800$$

Then,

The cost price of product C = $c = \text{Rs. } 800$

The cost price of product B = $6 * 800 = \text{Rs. } 4800$

The selling price of product A = $4800 - 1604 = \text{Rs. } 3196$

The selling price of product B = $6.3 * 800 = \text{Rs. } 5040$

The selling price of product C = $5040 * (495/2520) = \text{Rs. } 990$

The discount percentage on product A = $(x - 10)\% = 15\%$

The marked price on product A = $3196 * (100/85) = \text{Rs. } 3760$

The marked-up percentage of products B and C is '2x'% and '5y'% , respectively

So, the marked-up percentage of the products B and C is 50% and 65%, respectively.

The marked price on product B = $(150/100) * 4800 = \text{Rs. } 7200$

The marked price on product C = 165% of 800 = Rs. 1320

The profit price on product B = $5040 - 4800 = \text{Rs. } 240$

The profit price on product C = $990 - 800 = \text{Rs. } 190$

The discount price on product A = $3760 - 3196 = \text{Rs. } 564$

The discount price on product B = $7200 - 5040 = \text{Rs. } 2160$

The discount price on product C = $1320 - 990 = \text{Rs. } 330$

Let table,

Products	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	-	3760	3196	-	564
B	4800	7200	5040	240	2160
C	800	1320	990	190	330

Answer: B

The cost price of product A = $3196 \times (100/159.8) = \text{Rs. } 2000$

The marked-up percentage on product A = $[(3760 - 2000)/2000] \times 100 = (1760/2000) \times 100 = 88\%$

Thus, ratio = $88\% : 50\% = 44 : 25$

9. Questions

According to the Note,

$$x + y = 38 \rightarrow (1)$$

$$x^2 - y^2 = 456$$

$$(x + y)(x - y) = 456$$

$$38 * (x - y) = 456$$

$$x - y = 12 \rightarrow (2)$$

Solve (1) and (2)

$$2x = 50$$

$$x = 25$$

and

$$y = 38 - 25$$

$$y = 13$$

Let the cost price of product C be Rs. c

The cost price of product B = $6 * c = \text{Rs. } 6c$

The selling price of product A = Rs. $(6c - 1604)$

The profit percentage on product B = $(x/5)\% = (25/5)\% = 5\%$

The selling price of product B = $(105/100) * 6c = 6.3c$

The ratio of the selling prices of products A, B, and C is 1598: 2520: 495

According to the data,

$$(6c - 1604)/6.3c = 1598/2520$$

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$$(6c - 1604)/c = 799/200$$

$$1200c - 320800 = 799c$$

$$401c = 320800$$

$$c = 800$$

Then,

The cost price of product C = $c = \text{Rs. } 800$

The cost price of product B = $6 * 800 = \text{Rs. } 4800$

The selling price of product A = $4800 - 1604 = \text{Rs. } 3196$

The selling price of product B = $6.3 * 800 = \text{Rs. } 5040$

The selling price of product C = $5040 * (495/2520) = \text{Rs. } 990$

The discount percentage on product A = $(x - 10)\% = 15\%$

The marked price on product A = $3196 * (100/85) = \text{Rs. } 3760$

The marked-up percentage of products B and C is '2x%' and '5y%', respectively

So, the marked-up percentage of the products B and C is 50% and 65%, respectively.

The marked price on product B = $(150/100) * 4800 = \text{Rs. } 7200$

The marked price on product C = $165\% \text{ of } 800 = \text{Rs. } 1320$

The profit price on product B = $5040 - 4800 = \text{Rs. } 240$

The profit price on product C = $990 - 800 = \text{Rs. } 190$

The discount price on product A = $3760 - 3196 = \text{Rs. } 564$

The discount price on product B = $7200 - 5040 = \text{Rs. } 2160$

The discount price on product C = $1320 - 990 = \text{Rs. } 330$

Let table,

Products	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	-	3760	3196	-	564
B	4800	7200	5040	240	2160
C	800	1320	990	190	330

Answer: D

The cost price of product D = $4800 + (xy - 125) = 4675 + (325 - 125) = \text{Rs. } 5000$

The marked-up percentage on product D = $65\% * (6/5) = 78\%$

The discount percentage on product C = $[(1320 - 990)/1320] * 100 = (330/1320) * 100 = 25\%$

The discount percentage on product D = $25\% * (8/5) = 40\%$

The selling price of product D = $5000 * (178/100) * (60/100) = \text{Rs. } 5340$

Thus, average = $(5040 + 990 + 5340)/4 = 11370/4 = \text{Rs. } 2842.5$

10. Questions

According to the Note,

$$x + y = 38 \rightarrow (1)$$

$$x^2 - y^2 = 456$$

$$(x + y)(x - y) = 456$$

$$38 * (x - y) = 456$$

$$x - y = 12 \rightarrow (2)$$

Solve (1) and (2)

$$2x = 50$$

$$x = 25$$

and

$$y = 38 - 25$$

$$y = 13$$

Let the cost price of product C be Rs. c

The cost price of product B = $6 * c = \text{Rs. } 6c$

The selling price of product A = Rs. $(6c - 1604)$

The profit percentage on product B = $(x/5)\% = (25/5)\% = 5\%$

The selling price of product B = $(105/100) * 6c = 6.3c$

The ratio of the selling prices of products A, B, and C is 1598: 2520: 495

According to the data,

$$(6c - 1604)/6.3c = 1598/2520$$

$$(6c - 1604)/c = 799/200$$

$$1200c - 320800 = 799c$$

$$401c = 320800$$

$$c = 800$$

Then,

The cost price of product C = $c = \text{Rs. } 800$

The cost price of product B = $6 * 800 = \text{Rs. } 4800$

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The selling price of product A = $4800 - 1604 = \text{Rs. } 3196$

The selling price of product B = $6.3 * 800 = \text{Rs. } 5040$

The selling price of product C = $5040 * (495/2520) = \text{Rs. } 990$

The discount percentage on product A = $(x - 10)\% = 15\%$

The marked price on product A = $3196 * (100/85) = \text{Rs. } 3760$

The marked-up percentage of products B and C is '2x%' and '5y%', respectively

So, the marked-up percentage of the products B and C is 50% and 65%, respectively.

The marked price on product B = $(150/100) * 4800 = \text{Rs. } 7200$

The marked price on product C = 165% of 800 = Rs. 1320

The profit price on product B = $5040 - 4800 = \text{Rs. } 240$

The profit price on product C = $990 - 800 = \text{Rs. } 190$

The discount price on product A = $3760 - 3196 = \text{Rs. } 564$

The discount price on product B = $7200 - 5040 = \text{Rs. } 2160$

The discount price on product C = $1320 - 990 = \text{Rs. } 330$

Let table,

Products	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit price (Rs.)	The discount price (Rs.)
A	-	3760	3196	-	564
B	4800	7200	5040	240	2160
C	800	1320	990	190	330

Answer: E

The cost price of product A = $3 * 800 = \text{Rs. } 2400$

The profit price of product A = $3196 - 2400 = \text{Rs. } 796$

The sum of the profit prices on products A, B, and C together = $796 + 190 + 240 = \text{Rs. } 1226$

11. Questions

Answer: C

Quantity I,

$$(343x^3)^{(1/3)} - 3x^4 + 67x$$

$$7x - 3x^4 + 67x$$

$$74x - 3x^4$$

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Then,

$$3 \geq x \geq 2.$$

The possible values of x are 3 and 2,

If $x = 3$

Let,

$$= 74 * 3 - 3 * 3^4$$

$$= 222 - 243$$

$$= -21$$

If $x = 2$

Let,

$$= 74 * 2 - 3 * 2^4$$

$$= 148 - 48$$

$$= 100$$

Quantity II:

$$4 > y \geq 2$$

The possible values of y are 2 and 3,

If $y = 3$

Let,

$$= 24 * (729)^{(1/3)} - 31 * 3$$

$$= 24 * 9 - 93$$

$$= 123$$

If $y = 2$

Let,

$$= 24 * (729)^{(1/2)} - 31 * 2$$

$$= 24 * 27 - 62$$

$$= 586$$

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Quantity I	relation	Quantity II
100	<	123
100	<	586
-21	<	123
-21	<	586

Thus, Quantity I < Quantity II

Hence, option C

12. Questions

Answer: A

The area of rectangle = $l * b \text{ m}^2$

$$l * b = 234 \rightarrow (1)$$

In rectangle (The length is always higher than the breadth)

$$l - b = 5$$

$$l = b + 5 \rightarrow (2)$$

Equation (2) is substituted in (1)

$$(b + 5) * b = 234$$

$$b^2 + 5b = 234$$

$$b^2 + 5b - 234 = 0$$

$$b^2 + 18b - 13b - 234 = 0$$

$$b(b + 18) - 13(b + 18) = 0$$

$$(b + 18)(b - 13) = 0$$

$$b = 13 \text{ and } -18$$

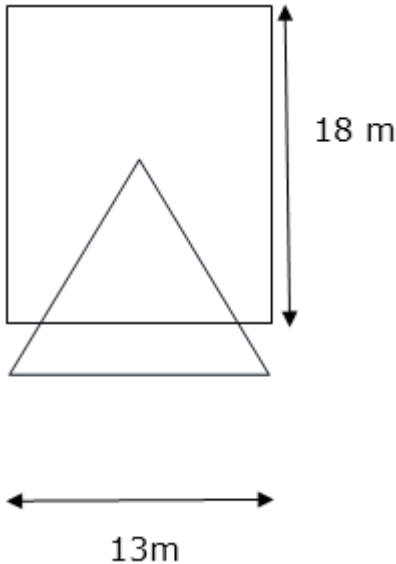
$$b = 13 \text{ (positive value)}$$

From (2)

$$l = 13 + 5$$

$$l = 18 \text{ m}$$

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So, the breadth of the rectangle is same as side of the equilateral triangle.

The equilateral triangle area $= (\sqrt{3}d^2/4) \text{ m}^2$

$d = \text{side}$

$d = 13$

Quantity I:

The value of ' d^2 ' $= 13 * 13 = 169$

Quantity II:

The side of the square $= 300\% \text{ of } 13 = 39 \text{ m}$

The area of square $= 39^2 = 1521$

Thus, Quantity I $<$ Quantity II

Hence, option A

13. Questions

Answer: B

The speed of the thief $= 43.2 * (5/18) = 12 \text{ m/sec}$

The distance travelled by thief in $66(2/3) \text{ sec} = 12 * (200/3) = 800 \text{ m}$ {Note: $66(2/3) = (200/3)$ }

The speed of police $= 's' \text{ m/sec}$

The relative speed between police and thief $= s - 12 \text{ m/sec}$

The time taken by police caught him in seconds $= 1 \text{ min } 40 \text{ seconds} = 100 \text{ seconds}$

According to the question,

$$800 = (s - 12) * 100$$

$$s - 12 = 8$$

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$$s = 20 \text{ m/sec}$$

The speed of police = 20 m/sec

Quantity I:

The speed of the boat = 20 m/s

The speed of the boat = $20 * 18/5 = 72 \text{ kmph}$

The speed of the stream = 4 km/hr

Thus, the time taken = $380 / (72 + 4) = 380/76 = 5 \text{ hours}$

Quantity II:

The speed of the police in km/hr = $20 * (18/5) = 72 \text{ km/hr}$

The time taken by the police to cover the distance 360 km = $360 / 72 = 5 \text{ hours}$

Thus, Quantity I = Quantity II

Hence, option B

14. Questions

Answer: C

Let the age of A, before 8 years and age of C before 5 years be $2x$ years and $5x$ years respectively.

The present age of A = $(2x + 8)$ years

The present age of C = $(5x + 5)$ years

The present age of B = $(3/1) * (2x + 8) = (6x + 24)$ years

According to the question,

$$6x + 24 + 6 = 2 * (5x + 5 - 10)$$

$$6x + 30 = 10x - 10$$

$$4x = 40$$

$$x = 10$$

The present age of A = $(2x + 8) = 28$ years

The present age of C = $(5x + 5) = 55$ years

The present age of B = $(6x + 24) = 60 + 24 = 84$ years

Here, $x = 28$, $y = 84$ and $z = 55$

Quantity I:

The sum of B, C, and D = $3 * (55 + 2) = 171$ years,

The present age of D = $171 - 84 - 55 = 32$ years

The age of A, before 6 years = $28 - 6 = 22$ years

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The required difference = $32 - 22 = 10$

Quantity II:

$$(x + z) * 10 \div y = (28 + 55) * 10 \div 84$$

$$= 830/84$$

$$= 415/42$$

$$= 9 (37/42)$$

Thus, Quantity I > Quantity II

Hence, option C

15. Questions

Answer: A

Formula

The difference between the simple interest and compound interest for 2 years = $PR^2/100^2$

The difference between the simple interest and compound interest for 3 years = $PR^2 * (300 + R)/100^3$

Simple interest = $PNR/100$

According to the question,

$$[PR^2/100^2]/[PR^2 * (300 + R)/100^3] = 10/31$$

$$100/(300 + R) = 10/31$$

$$310 = 300 + R$$

$$R = 10$$

Then,

$$P * 1 * (10/100) = 2500$$

$$P = \text{Rs. } 25000$$

Quantity I:

$$\text{Principle} = P + 2250 = 25000 + 2250 = \text{Rs. } 27250$$

$$R = 9\%$$

$$\text{Interest received} = 27250 * 9 * (4/100) = \text{Rs. } 9810$$

Quantity II:

$$\text{Principle} = P - 5000 = 25000 - 5000 = \text{Rs. } 20000$$

$$R = 20\%$$

$$\text{Interest received} = 20000 * [120/100]^3 - 20000$$

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$$= 20000 * \{(6/5)^3 - 1\}$$

$$= 20000 * [216 - 125]/125$$

$$= \text{Rs. } 14560$$

Thus, Quantity I < Quantity II.

Hence, option A

16. Questions

Answer: C

The given series follows the following pattern:

$$243 * 2 = 486$$

$$486/3 = 162$$

$$162 * 2 = \mathbf{324}$$

$$324/3 = 108$$

$$108 * 2 = 216$$

$$216/3 = 72$$

So, the wrong and correct term of the series is 327 and 324, respectively.

$$p = 327$$

$$q = 324$$

$$p + q = 327 + 324 = 651$$

So, 648 is the nearest value.

17. Questions

Answer: A

The given series follows the following pattern:

$$X - (X - 7) = 7$$

$$95 - 76 = 19$$

$$123 - 95 = 28$$

$$162 - 123 = 39$$

So, the series formed by difference of difference

57	64	76	95	123	162
7	12	19	28	39	
5	7	9	11		

So, X = 64

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Let,

The new series is started with $\sqrt{X}$

$$\sqrt{64} = 8$$

The finding term = $\sqrt[3]{X} = \sqrt[3]{64} = 4$

Let,

8	15	27	46
7	12	19	
5	7		

So, the fourth term is 46.

18. Questions

Answer: E

$$y = \{1 + (64 - 14)/100\} \text{ of } 64 = (150/100) * 64 = 96$$

The first term of the series is decreased by 3 = 57 - 3 = 54

$$\text{Thus, percentage} = (54/96) * 100 = 56.25\%$$

19. Questions

Answer: D

From series II,

The given series follows the following pattern:

$$0.04 * 12.5 = 0.5$$

$$0.5 * 25 = 12.5$$

$$12.5 * 50 = 625$$

$$625 * 100 = 62500$$

So,

$$500q = 62500$$

$$q = 125$$

Then series I,

$$500q, 2500, (q - 25), p, 0.16$$

$$62500, 2500, 100, p, 0.16$$

The given series follows the following pattern:

$$62500/25 = 2500$$

$$2500/25 = 100 \quad (q - 25 = 100)$$

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$$100/25 = 4$$

$$4/25 = 0.16$$

$$\text{So, } p = 4$$

$$\text{The required value} = q/(p + 1) = 125/5 = 25$$

20. Questions

Answer: C

Series: 4, 8, 24, ?, 480, 2880

The given series follows the following pattern:

$$4 * 2 = 8$$

$$8 * 3 = 24$$

$$24 * 4 = \mathbf{96}$$

$$96 * 5 = 480$$

$$480 * 6 = 2880$$

So, the missing term is 96.

Hence, option C

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